

Mock It Till You Make It

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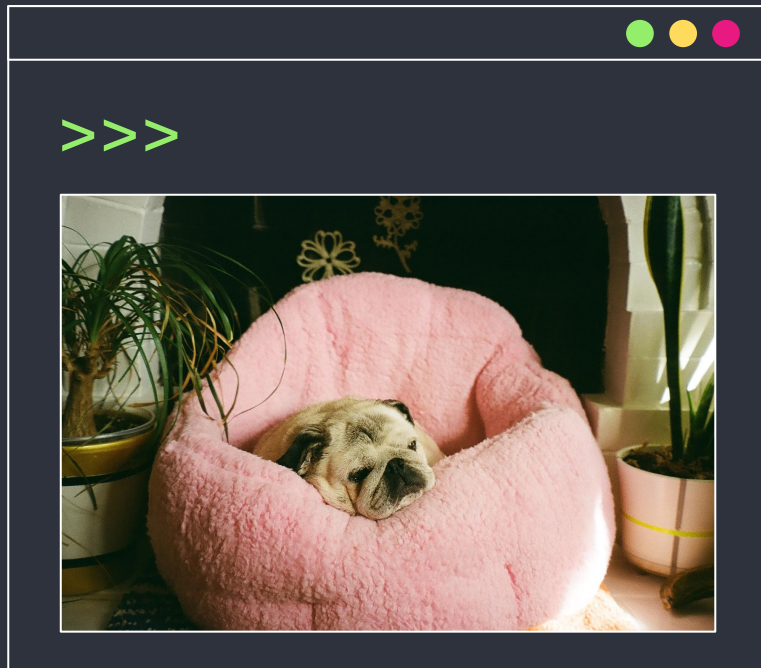
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How to Verify Your
External Mocks Without
Ever Leaving Unittest

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</ Hi, my name is Liz! [she/her/hers]

- Film student → Marketer → Engineer → Developer advocate → Developer content marketing manager
- I love pizza, plants, pugs, and Python
- Tech ⇌ Community
- Complex concepts ⇌ Casual conversation



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</ Agenda

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Testing and its benefits

{02}

Unit testing

{03}

Mocking

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Demo time!

{05}

Q&A

</ What is software testing?



Software testing is the process of **evaluating** and **verifying** that a software product or application **does what it is supposed to do**.¹

¹“What is software testing?” <https://www.ibm.com/topics/software-testing>

</ Testing and its benefits



Prevent
bugs



Accelerate
development



Save
money

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</ Testing and its benefits



Write **better** code!

- Detect code smells
- Easier to understand, maintain, and debug

</ Testing and its benefits

	GOOD	NEUTRAL	EVIL
LAWFUL	ACCEPTANCE TESTING Does the whole system work as intended? 	UNIT TESTING Do the smallest components of the system work correctly independently? 	REGRESSION TESTING Do new features break or degrade functionality? 
NEUTRAL	INTEGRATION TESTING  Does the different software components or functions operate together?	USABILITY TESTING How well can a customer use a system or web application to complete a task? 	FUNCTIONAL TESTING Does the system meet the requirements of the intended business scenarios? 
CHAOTIC	STRESS TESTING  How much strain can the system endure before it fails?	PERFORMANCE TESTING How does the system perform under different workloads? 	CHAOS TESTING Can the system withstand unpredictable, turbulent conditions? 

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</ What is unit testing?



Unit testing is the process of *evaluating* and *verifying* that the *smallest functional units* of code do what they are supposed to do *individually* and *independently*.

</ Unit testing: Best practices



One precise
assertion at a
time

One precise assertion at a time

Independent, isolated, and
controlled

Relevant and meaningful

Repeatable and deterministic

Automatic

Descriptive

</ Unit testing: Best practices



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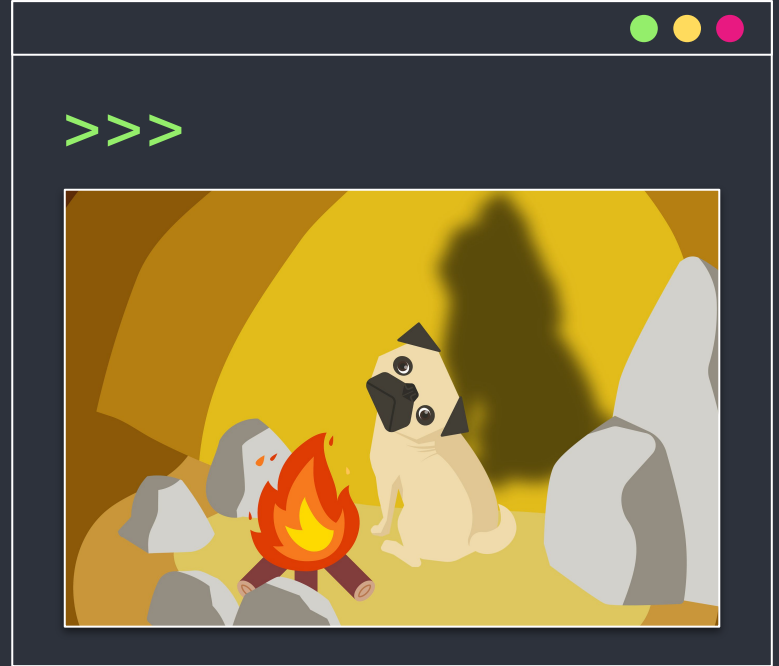
Repeatable and deterministic

Automatic

Descriptive

</ What is mocking?

- Testing tool
- Temporarily replace a method, dependencies, or behavior
- Create a more controlled test environment
- Limit dependencies, eliminate additional costs, and prevent vulnerabilities



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</ What is mocking?

- Additional code to maintain
- Fall out of sync with reality
- Render unit tests irrelevant and meaningless



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</ Mocking: Best practices



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Use a mocking library

Mock only what you need

Verify mocks for accuracy

Keep mocks simple

Refactor your code

</ Mocking: Best practices



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</ unittest.mock

Mock and MagicMock

```
mocked_pug_facts_response =  
MagicMock(spec=requests.Response)
```

```
mock_requests.get.return_value  
= mocked_pug_facts_response
```

patch[]

```
@patch('pug.requests',  
autospec=True)  
def test_get_pug_facts(self,  
mock_requests):
```


</ unittest.skip

`@unittest.skip(reason)`

`@unittest.skip(reason)`

Skipping on conditionals

`@unittest.skipIf(condition,
reason)`

`@unittest.skipUnless(condition
, reason)`

</ Demo time!

Build-a-Pug Pug Facts

Welcome to Build-a-Pug!

Fill in the form to build your own pug!

Name

Age

Home

Puppy Dinner

Submit



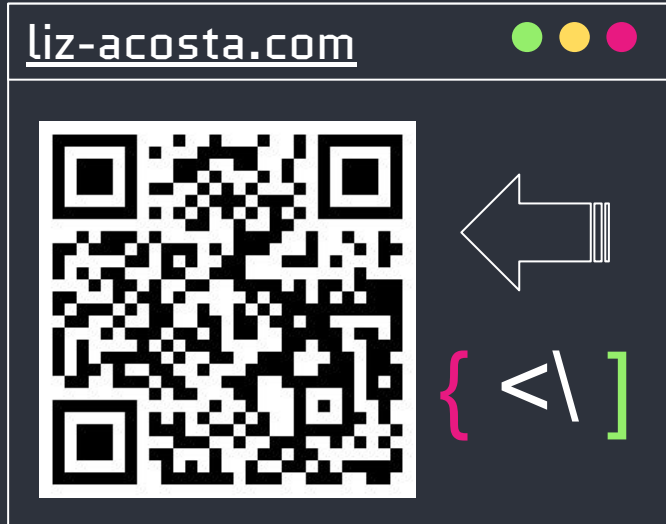
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Thank you!!!

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</ Questions?



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